REAL-WORLD PROBLEM SOLVING WITH POWERSHELL

Class Presentation Report

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1. Abstract This project demonstrates real-world cybersecurity investigation using Windows PowerShell. Eight practical tasks were completed to simulate forensic analysis of a Windows endpoint using only built-in PowerShell tools. 2. Introduction: Why PowerShell is Important in Cybersecurity PowerShell is critical for cybersecurity investigations because it has direct access to Windows OS objects, registry, event logs, and processes without needing external tools. It can automate forensics, query WMI/CIM data, and run remotely via WinRM. Attackers abuse PowerShell for fileless malware, so defenders must master it to detect abuse. 3. Methodology Tool: Windows PowerShell run as Administrator Method: Each task was executed with native cmdlets. -ErrorAction SilentlyContinue was used to skip access-denied folders. Practical Tasks & PowerShell Solutions Tasks 1–8 and analyses included as provided. REAL-WORLD PROBLEM QUESTIONS - PowerShell Answers Questions 1–5 and answers included as provided. POWER SHELL PRACTICAL CHALLENGES Challenge 1: List all hidden files in the system Get-ChildItem -Path C:\ -Force -Recurse -File -ErrorAction SilentlyContinue | Where-Object {$_.Attributes -match "Hidden"} Challenge 2: Find all .exe files inside user directories only Get-ChildItem C:\Users -Recurse -Filter *.exe -ErrorAction SilentlyContinue Challenge 3: Show files modified in the last 7 days Get-ChildItem C:\Users -Recurse -File | Where-Object {$_.LastWriteTime -gt (Get-Date).AddDays(-7)} Challenge 4: List the top 10 largest files in C:\Users Get-ChildItem C:\Users -Recurse -File -ErrorAction SilentlyContinue | Sort-Object Length -Descending | Select-Object FullName, Length -First 10 Challenge 5: Scan AppData for suspicious executable files and export results to a text file Get-ChildItem $env:APPDATA -Recurse -Filter *.exe | Select-Object FullName, Length, LastWriteTime | Out-File -FilePath C:\suspicious_exe.txt Challenges Faced 1. Access Denied: Solved with -ErrorAction SilentlyContinue 2. Long scan times: Limited searches to C:\Users instead of full C:\ 3. Documentation: Used Export-Csv / Out-File to save evidence Conclusion PowerShell alone is powerful for initial Windows endpoint forensics. Recommendation Enable PowerShell Script Block Logging and use Get-FileHash + VirusTotal for all unknown files. Appendixes Later...